

Assignment #5

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Overview:

In this Assignment you will create Go programs to connect microservices to a Postgres database server that is locally installed as an application on the personal development computer. You will use psql tool to inspect the data injected by the microservice into the database tables hosted on the Postgres server.

Submission:

Submit your exercise as a SINGLE ZIP file on Canvas by the due date. Your submitted ZIP file must have the name:
Assignment_5_Your_LastName.zip

Deliverables:

Your ZIP file for the Assignment submission must include the following:

- All source code that you wrote, installed, compiled, and built on your personal development computer.
- Panopto video recording of a live run (5 to 10 minutes) of your code on your personal development computer.

Requirements Specification:

Implement the following requirements, and test-drive your Go application:

1. (10 Points) Build Docker/Container Postgres image by executing the following command:

```
docker-compose up --build
```

2. (10 Points) Use **psql** to execute the following command from the command window/terminal to create the go database and Users and UserData tables:

```
"C:\Program Files\PostgreSQL\13\bin\psql" -h localhost -p 5433 -U postgres < create_tables.sql
```

3. (10 Points) Run the following Go program 5 times:

```
go run postGO.go
```

4. (10 Points) Use **psql** to execute the following commands from the command window/terminal to inspect the tables in the go database and Users and UserData tables:

```
4.1. "C:\Program Files\PostgreSQL\13\bin\psql" -h localhost -p 5433 -U postgres
```

```
4.2. \l
```

```
4.3. \c go
```

```
4.4. \dt
```

```
4.5. select * from userdata;
```

5. (10 Points) Modify the provided Go programs to create a **database** with the name **MSDS** that has the **table** with the name **MSDSCourseCatalog** that has 5 MSDS courses entries each has the following MSDSCourse structure:

```
type MSDSCourse struct {  
    CID string `json:"course_D`  
    CNAME string `json:"course_name"`  
    CPREREQ string `json:"prerequisite"`  
}
```

6. (10 Points) Use **psql** commands to demonstrate the content of the MSDSCourseCatalog table after your run.